

PRODUCT REQUIREMENTS DOCUMENT (PRD)

Data Analytics & Reporting Platform

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1. Executive Summary

Product Name

Data Analytics & Reporting Platform

Product Vision

To provide organizations with a self-service analytics platform that transforms raw business data into actionable insights through automated data processing, interactive dashboards, and intelligent reporting.

Product Mission

Enable users with no technical background to upload datasets, analyze information, visualize trends, and make informed decisions.

Product Category

Business Intelligence (BI) and Analytics Software

2. Problem Statement

Organizations collect large volumes of business data from:

- Sales Systems
- CRM Systems
- ERP Systems
- Marketing Platforms
- Excel Sheets

Most organizations face:

- Poor visibility into business performance
- Time-consuming manual reporting
- Data inconsistencies
- Delayed decision-making
- Lack of centralized analytics

Current reporting processes often require technical expertise and significant manual effort.

3. Proposed Solution

Develop a cloud-based analytics platform that:

1. Accepts datasets in CSV or Excel format
2. Automatically validates uploaded data
3. Cleans and preprocesses records
4. Performs statistical analysis
5. Generates visualizations
6. Creates dashboards
7. Produces downloadable reports

The platform reduces analysis time and improves business intelligence capabilities.

4. Business Goals

Goal 1

Reduce manual reporting efforts by 80%.

Goal 2

Enable data-driven decision making.

Goal 3

Improve reporting accuracy.

Goal 4

Provide real-time business visibility.

Goal 5

Increase productivity of analysts and managers.

5. Target Users

Business Managers

Need performance monitoring.

Data Analysts

Need advanced analytical capabilities.

Executives

Need KPI dashboards.

Sales Teams

Need sales trend analysis.

Operations Teams

Need operational reporting.

6. User Personas

Persona 1

Business Manager

Pain Points:

- Delayed reports
- Manual Excel work
- Lack of insights

Goals:

- Monitor KPIs quickly
 - Track sales performance
-

Persona 2

Data Analyst

Pain Points:

- Data cleaning effort
- Multiple data sources

Goals:

- Faster analysis
 - Better visualizations
-

7. Product Features

Feature 1

Dataset Upload

Description

Allows users to upload:

- CSV Files
- XLS Files
- XLSX Files

Acceptance Criteria

- Upload success under 5 seconds
 - File validation completed automatically
-

Feature 2

Automated Data Validation

Checks

- Missing Values
 - Invalid Datatypes
 - Duplicate Records
 - Empty Columns
 - Invalid Dates
-

Feature 3

Data Cleaning Engine

Functions

- Remove duplicates
 - Handle missing values
 - Format dates
 - Normalize columns
 - Detect outliers
-

Feature 4

Analytics Engine

Capabilities

- Descriptive Statistics
 - Correlation Analysis
 - Trend Analysis
 - Category Analysis
 - Customer Analysis
-

Feature 5

Visualization Engine

Charts

- Line Chart
 - Bar Chart
 - Pie Chart
 - Histogram
 - Scatter Plot
 - Heatmap
-

Feature 6

Dashboard Builder

Components

- KPI Cards
 - Interactive Charts
 - Filters
 - Drill-down Reports
-

Feature 7

Report Export

Formats

- PDF
 - Excel
 - CSV
-

8. Functional Requirements

FR-001

User shall upload datasets.

FR-002

System shall validate file format.

FR-003

System shall clean uploaded data.

FR-004

System shall calculate statistics.

FR-005

System shall generate dashboards.

FR-006

System shall export reports.

FR-007

System shall maintain user sessions.

FR-008

System shall provide filters.

FR-009

System shall store historical reports.

FR-010

System shall support role-based access.

9. Non-Functional Requirements

Performance:

Dashboard loads under 3 seconds.

Scalability:

Support 100,000+ records.

Availability:

99.9% uptime.

Security:

JWT Authentication.

Reliability:

Daily backups.

Accessibility:

WCAG Compliance.

10. UI/UX Requirements

Design Principles

- Simplicity
- Consistency
- Accessibility
- Responsiveness

Dashboard Layout

Header

Sidebar

KPI Section

Charts Section

Reports Section

Footer

11. Wireframe

HEADER

Search

Notifications

Profile

SIDEBAR

Dashboard

Reports

Analytics

Settings

MAIN

KPI Cards

Sales Trend

Profit Trend

Category Performance

Regional Analysis

Customer Analysis

12. Frontend Architecture

Framework:

React.js

Styling:

Tailwind CSS

Visualization:

Chart.js

State Management:

Redux

Routing:

React Router

13. Backend Architecture

Framework:

FastAPI

Language:

Python

Modules:

Authentication Service

Analytics Service

Reporting Service

Data Processing Service

Notification Service

14. Database Design

Users

Reports

Datasets

DashboardConfigurations

AnalyticsResults

AuditLogs

15. API Requirements

POST /upload

POST /login

GET /dashboard

GET /analytics

GET /reports

POST /export

DELETE /dataset

16. Security Requirements

JWT Authentication

Password Hashing

Role-Based Access

Input Validation

SQL Injection Prevention

HTTPS Encryption

Audit Logging

17. Analytics Engine Logic

Step 1

Load Dataset

Step 2

Validate Dataset

Step 3

Clean Data

Step 4

Transform Data

Step 5

Generate Statistics

Step 6

Create Visualizations

Step 7

Generate Insights

Step 8

Display Dashboard

18. Deployment Architecture

Frontend

Vercel

Backend

Render

Database

MySQL

Storage

AWS S3

Monitoring

Grafana

Logging

ELK Stack

19. Success Metrics

Monthly Active Users

Dashboard Usage Rate

Report Export Rate

Upload Success Rate

System Availability

Average Load Time

20. Future Roadmap

Phase 2

Machine Learning Forecasting

Phase 3

AI Generated Insights

Phase 4

Natural Language Queries

Phase 5

Real-Time Streaming Analytics

21. MVP Scope

Dataset Upload

Data Cleaning

EDA

Dashboard

Report Export

Authentication

Basic Analytics

22. Risks

Large File Uploads

Data Quality Issues

Performance Bottlenecks

Security Vulnerabilities

Third-Party Service Failures

23. Acceptance Criteria

All core modules operational.

Dashboard generation successful.

Data cleaning accuracy above 95%.

System uptime above 99%.

Report exports functioning correctly.

24. Conclusion

The Data Analytics & Reporting Platform will provide organizations with an end-to-end solution for transforming raw business data into actionable intelligence through automated analytics and reporting.